NO.4 i-stop FUNCTION DOES NOT OPERATE [SKYACTIV-D 1.5]

id1103q4000900

4	i-stop FUNCTION DOES NOT OPERATE
DESCRIPTION	• i-stop function does not operate when vehicle is stopped.
POSSIBLE CAUSE	• The i-stop (engine stop control) permit conditions are not met due to either of the following causes: — Driver operation — Environment (problem other than with vehicle) — i-stop related part malfunction **Note** • PIDs related to i-stop can be displayed in a list using the M-MDS data monitor diagnostic function. • If the engine is stopped by the i-stop control while the following data monitor items on the M-MDS are being viewed, the data monitor items may no longer be displayed. — BRAKE_SW — BRK_F_P_C — LON_ACCL_C — WSPD_SEN_LF — WSPD_SEN_RF — WSPD_SEN_LR — WSPD_SEN_RR

i-stop (engine-stop control) permit condition

Determination module	Determination item	Determination condition		Applicable PID
		ATX	MTX	
PCM	Vehicle speed	0 km/h {0 mph}	3 km/h {2 mph} or less	WSPD_SEN_LF WSPD_SEN_RF WSPD_SEN_LR WSPD_SEN_RR
	Brake pedal	Brake pedal is depressed while in D or M position	Not applicable	TR BOO, BPA
	Brake fluid pressure	Brake fluid pressure is 1.7 MPa {17 kgf/cm^2, 247 psi} or more while in D or M position (perform correction according to vehicle inclination angle)	Not applicable	TR BRK_F_P_C
	Accelerator pedal	Released	←	APP
	Clutch pedal	Not applicable	Released	CPP
	MTX gear position	Not applicable	Neutral	CPP/PNP
	Vehicle conditions	Vehicle is stopped in D or M position*1, *8	Not applicable	TR
	Steering speed	15 deg/s or less	←	STR_ROT_SPD
	Steering angle	Within 65° of center to left and right (After EPS control module learns center value)	Not applicable	STR_AB_EST
	Steering torque	1.1 N·m {11 kgf·cm, 9.7 in·lbf} or less	←	STR_TRQ_S_M
	i-stop OFF switch	OFF	←	I-Stop_OFF
	Vehicle speed history (After engine started and driven once)	3 km/h {2 mph} or more	4 km/h {2.5 mph} or more	I-Stop_VSP
	i-ELOOP bypass mode operation condition	Bypass mode is not operated (BYPS_STAT: Bypass Off)	←	BYPS_STAT
	Capacitor (i-ELOOP) voltage	14 V or more	←	DCDC_IN_V
	Battery charge condition	65.7 % or more	←	BATT_SOC
	Battery fluid temperature	0—70 °C {32—158 °F}	←	BATT_TEMP
	Battery voltage	11.2 V or more	←	BATT_V
	Power brake unit vacuum	−45 kPa {−0.46 kgf/cm^2, −6.5 psi} or less (value calculated by subtracting BARO from BBP)	←	BBP BARO
	Door (driver's side)	Closed*3	←	DOOR_D
	Bonnet	Closed*3	←	HOOD

Determination module	Determination item	Determination condition		Applicable PID
		ATX	MTX	
PCM	Vehicle inclination angle	Within ±7° when level (LON_ACCL_C: within ±0.121 G)	Not applicable	LON_ACCL_C
	Seat belt (driver's side)	Fastened *4	←	SEAT_B_D
	Fuel injection amount learning	Not implemented*5	←	Not applicable
	Diesel particulate filter regeneration	Not implemented	←	Not applicable
	Battery condition setting	Completed	←	Not applicable
	Steering angle sensor initial setting	Completed	Not applicable	Not applicable
	Engine coolant temperature	30—110 °C {86—230 °F}	←	ECT
	Intake air temperature	100 °C {212 °F} or less	←	IAT
	Altitude	1,800 m or less (BARO: 81.5 kPa {0.831 kgf/cm^2, 11.8 psi} or more)	←	BARO
	System condition	i-stop related part DTCs are not recorded	←	Not applicable
Start stop unit	+B (ROOM fuse) input	Yes	←	VPWR_B1
	IG1 input	Yes	←	VPWR_IG1
	ACC input	Yes	←	VPWR_ACC
	IG1 output	Yes	←	RELAY_IG1
	ACC output	Yes	←	RELAY_ACC
	Push button	Not pressed	←	SSB_1
				SSB_2
	Door (all)	Closed	←	DOOR_ALL
	Trunk (4SD)/Liftgate (5HB)	Closed	←	TRUNK
	Seat belt (driver's side)	Fastened *4	←	SEAT_B_D
	Engine start	Engine was started by operating push button with key inside cabin (engine started first time)	←	Not applicable
	Engine rotation speed	300 rpm or more is maintained after engine started	←	Not applicable
	System condition	i-stop related part DTCs are not recorded	←	Not applicable
TCM	TCM detection condition	All met (I-Stop_EST: On)	Not applicable	I-Stop_EST
	Panic stop	Brake pedal was not depressed suddenly within 4 s	Not applicable	Not applicable
	TFT	20—117 °C {68—242 °F}	Not applicable	TFT
	Gear position	Not in 2nd gear fixed mode	Not applicable	Not applicable
	System condition	i-stop related part DTCs are not recorded	Not applicable	Not applicable

Determination module	Determination item	Determination condition		Applicable PID
		ATX	MTX	
Climate control unit	Cabin temperature (with full-auto air conditioner system)	Difference between target cabin temperature and actual cabin temperature is a certain value or less (Cabin temperature control by A/C is normally performed)	←	Not applicable
	A/C temperature (with full-auto air conditioner system)	Setting other than MIN *6	←	Not applicable
	Ambient temperature	−10—50 °C {14—122 °F}	←	Not applicable
	Warm-up operation condition (with manual air conditioner system)	Ambient temperature is 10 °C {50 °F} or more or engine coolant temperature is 60 °C {140 ° F} or more	←	ECT
	Cold condition (with manual air conditioner system)	Ambient temperature is 20 °C {68 °F} or less or evaporator temperature is specified value or less	←	Not applicable
	MODE dial	Except DEFROSTER	←	Not applicable
	Airflow volume control dial (with full-auto air conditioner system)	OFF *7	←	Not applicable
	System condition	i-stop related part DTCs are not recorded	←	Not applicable
DSC HU/CM	Operation request from PCM	Hill Launch Assist (HLA) control request signal from PCM is received	Not applicable	Not applicable
	Wheel speed (4 wheels)	0 km/h {0 mph}	Not applicable	WSPD_SEN_LF
				WSPD_SEN_RF
				WSPD_SEN_LR
				WSPD_SEN_RR
	Brake pedal	Brake pedal is depressed	Not applicable	BRAKE_SW
	Brake fluid pressure	Approx. 0.45 MPa {4.6 kgf/cm^2, 65 psi} or more	Not applicable	BRK_F_P_C
	Accelerator pedal	Released	Not applicable	APP
	System condition	i-stop related part DTCs are not recorded	Not applicable	Not applicable

*1 : After the vehicle is stopped, the engine stops if the gear is shifted to N position. In addition, if the gear is shifted to P position after the vehicle is stopped in D position, the engine stop condition by the i-stop control is maintained.
*2 : If a door is opened while the vehicle is stopped, i-stop is inhibited until a vehicle speed record is input once..
*3 : If the engine is started with the bonnet open, i-stop is inhibited until the engine is stopped.
*4 : If the seat belt is unfastened while the vehicle is stopped, i-stop is inhibited until a vehicle speed record is input once.
*5 : If the fuel injection amount learning, which is periodically performed automatically, is not completed, the PCM turns the i-stop indicator light (green) off and inhibits i-stop when the vehicle speed is approx. 10 km/h {6.2 mph} or less before the vehicle stops.
*6 : When the A/C set temperature is set to MAX, the PCM may inhibit the i-stop control.
*7 : When all of the other i-stop permit conditions related to the climate control unit are met even if the airflow volume control dial is on, the i-stop control is permitted.
*8 : When shifting to the R position and driving the vehicle, the i-stop function is inhibited until the vehicle is driven at 5 km/h {3.0 mph} or more for 1 min or more after shifting to the D or M position.

Diagnostic Procedure

Step	Inspection	Results	Action
1	**VERIFY DTCs** • Using the M-MDS, display the DTCs related to the PCM, TCM, DSC HU/CM, climate control unit, and start stop unit. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See ON-BOARD DIAGNOSTIC SYSTEM DTC INSPECTION [TCM (EW6A-EL)].) (See DTC INSPECTION [DSC HU/CM].) (See DTC TABLE [FULL-AUTO AIR CONDITIONER].) (See DTC INSPECTION [START STOP UNIT].) • Is a DTC displayed?	Yes	Repair the malfunctioning location according to the applicable DTC troubleshooting. (See DTC TABLE [SKYACTIV-D 1.5].) (See ON-BOARD DIAGNOSTIC SYSTEM DTC TABLE [TCM (EW6A-EL)].) (See DTC TABLE [DSC HU/CM].) (See DTC TABLE [FULL-AUTO AIR CONDITIONER].) (See DTC TABLE [START STOP UNIT].)
		No	Go to the next step.
2	**VERIFY i-STOP (ENGINE-STOP CONTROL) PERMIT CONDITION ITEMS** **Warning** **• To prevent an accident, work with two people when the vehicle is being driven (one person drives vehicle, other person operates M-MDS).** **• Observe legal speed limits when driving the vehicle and perform the procedure in a location where safety can be assured.** • Switch the ignition ON (engine off). • Display the PID related to the i-stop (engine-stop control) permit conditions using the M-MDS. (See PID/DATA MONITOR INSPECTION (i-stop) [SKYACTIV-D 1.5].) • Drive the vehicle. • Verify if the i-stop (engine-stop control) permit conditions of the items which can be verified by the PID are met. • Are the permit conditions of all the items met?	Yes	Go to the next step.
		No	Inspect the applicable items for Inspection item(s).
3	**VERIFY VEHICLE CONDITION RELATED TO i-STOP (ENGINE-STOP CONTROL) PERMIT CONDITION** • Verify if all the following vehicle conditions are met. — Engine starts by operating push button with key left in vehicle — Brake pedal not depressed suddenly — Gear position: Not in 2nd gear fixed mode • Are all the conditions met?	Yes	Go to the next step.
		No	Verify the i-stop (engine-stop control) permit condition, then go to Step 11.
4	**PERFORM BATTERY CONDITION INITIAL SETTING (i-STOP SETTING)** • Perform the battery condition initial setting (i-stop setting). (See BATTERY CONDITION INITIALIZATION SETTING (i-stop SETTING).) • Drive the vehicle to verify the i-stop operation condition. • Is engine-stopping by i-stop operable?	Yes	Go to Step 11.
		No	Go to the next step.
5	**INSPECT AMBIENT TEMPERATURE SENSOR FOR MALFUNCTION** • Inspect the ambient temperature sensor. (See AMBIENT TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) (See AMBIENT TEMPERATURE SENSOR INSPECTION [MANUAL AIR CONDITIONER].) • Is the ambient temperature sensor normal?	Yes	Go to the next step.
		No	Replace the ambient temperature sensor, then go to Step 11. (See AMBIENT TEMPERATURE SENSOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].) (See AMBIENT TEMPERATURE SENSOR REMOVAL/INSTALLATION [MANUAL AIR CONDITIONER].)

Step	Inspection	Results	Action
6	**INSPECT EVAPORATOR TEMPERATURE SENSOR FOR MALFUNCTION** • Inspect the evaporator temperature sensor. (See EVAPORATOR TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) (See EVAPORATOR TEMPERATURE SENSOR INSPECTION [MANUAL AIR CONDITIONER].) • Is the evaporator temperature sensor normal?	Yes	Vehicles with full-auto A/C • Go to the next step. Vehicles with manual A/C • Go to Step 10.
		No	Replace the evaporator temperature sensor, then go to Step 11. (See EVAPORATOR TEMPERATURE SENSOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].) (See EVAPORATOR TEMPERATURE SENSOR REMOVAL/INSTALLATION [MANUAL AIR CONDITIONER].)
7	**INSPECT CABIN TEMPERATURE SENSOR FOR MALFUNCTION** • Inspect the cabin temperature sensor. (See CABIN TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is the cabin temperature sensor normal?	Yes	Go to the next step.
		No	Replace the cabin temperature sensor, then go to Step 11. (See CABIN TEMPERATURE SENSOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].)
8	**DETERMINE IF MALFUNCTION CAUSED BY AIR MIX ACTUATOR MALFUNCTION** • Inspect the air mix actuator. (See AIR MIX ACTUATOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is the air mix actuator normal?	Yes	Inspect the air mix door and linkage for sticking. (See A/C UNIT DISASSEMBLY/ASSEMBLY.) • If the steering switch is normal: — Go to the next step. • If there is any malfunction: — Repair or replace the malfunctioning location, then go to Step 11.
		No	Replace the air mix actuator, then go to Step 11. (See AIR MIX ACTUATOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].)
9	**DETERMINE IF MALFUNCTION CAUSED BY AIRFLOW MODE ACTUATOR MALFUNCTION** • Inspect the airflow mode actuator. (See AIRFLOW MODE ACTUATOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is the airflow mode actuator normal?	Yes	Go to the next step.
		No	Replace the airflow mode actuator, then go to Step 11. (See AIRFLOW MODE ACTUATOR REMOVAL/INSTALLATION [FULL-AUTO AIR CONDITIONER].)
10	**INSPECT EPS CONTROL MODULE FOR MALFUNCTION** • Inspect the EPS control module. (See EPS CONTROL MODULE INSPECTION.) • Is the EPS control module normal?	Yes	Perform the following procedures, then go to the next step. 1. Switch the ignition to OFF, and maintain this condition for approx. 3 s. 2. Set the vehicle wheels in the straight-ahead position. 3. Switch the ignition ON (engine off or on).
		No	Replace the EPS control module, then go to the next step. (See STEERING WHEEL AND COLUMN REMOVAL/INSTALLATION.)
11	**VERIFY THAT REPAIRS HAVE BEEN COMPLETED** • Has the malfunction symptom been eliminated?	Yes	Complete the symptom troubleshooting. (Explain contents of repair to customer)
		No	Repeat the inspection from Step 1.